Tenecteplase

Mechanism of action

- Tenecteplase is a fibrinolytic that accelerates the breakdown of blood clots. It converts the plasma protein plasminogen into the active enzyme plasmin, which breaks down fibrin within blood clots.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ STEMI when following the fibrinolytic pathway.

Contraindications

- ✕ Known severe allergy.

Tenecteplase

Cautions

- More than ten minutes of CPR.
- Non-compressible vascular puncture within the last 24 hours.
- Internal bleeding within the last six weeks.
- Lumbar puncture or epidural insertion within the last six weeks.
- TIA within the last three months.
- Known bleeding disorder.
- Taking an anticoagulant.
- Systolic BP greater than 180 mmHg or diastolic BP greater than 110 mmHg.
- Known to be pregnant or less than two weeks postpartum.
- Time of onset of symptoms was greater than 12 hours ago.
- Requires assistance with their activities of daily living.
- Severe comorbidities that significantly limit their life expectancy.
- Suspected aortic dissection.
- Major surgery, major trauma or severe brain injury within the last six weeks.
- Intracranial surgery within the last six months.
- Ischaemic stroke within the last six months.
- Previous intracerebral haemorrhage.
- Known cerebral aneurysm, arteriovenous malformation or tumour.

Use in pregnancy or when breastfeeding

- Administration during pregnancy or within two weeks of birth carries a significant risk of bleeding. The likelihood of STEMI occurring in a woman who is pregnant or within two weeks of birth is very low and personnel must seek clinical advice prior to administration.

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- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- Dosage is based on age and known (or estimated) weight.

Weight	Age less than 75 years		Age 75 years or older	
	Tenecteplase (dose IV)	Tenecteplase (volume IV)	Tenecteplase (dose IV)	Tenecteplase (volume IV)
<60 kg	30 mg	6 ml	15 mg	3 ml
60-69 kg	35 mg	7 ml	17.5 mg	3.5 ml
70-79 kg	40 mg	8 ml	20 mg	4 ml
80-89 kg	45 mg	9 ml	22.5 mg	4.5 ml
≥90 kg	50 mg	10 ml	25 mg	5 ml

Administration

- Dissolve the powder using the syringe within the kit.
- Carefully discard unwanted drug from the syringe, preferably into the ampoule before administration, ensuring the correct dose remains in the syringe.
- If an error is made in discarding unwanted drug and the correct dose cannot be drawn up, administer the remaining drug, document this well, and notify the receiving clinician.
- Administer undiluted as an IV bolus, preferably into a running IV line.

Common adverse effects

- Bleeding. Tenecteplase commonly causes superficial bleeding, including epistaxis, bruising and bleeding from IV sites.
- Dysrhythmia. It is common for dysrhythmia to occur if the coronary artery reperfuses. Most commonly the rhythm is accelerated idioventricular rhythm (AIVR) which does not require specific treatment. Other dysrhythmias should be treated using the appropriate guideline.

Tenecteplase

Usual onset of effect

- 5-10 minutes.

Usual duration of effect

- 2-6 hours.

Pharmacokinetics

- Tenecteplase is metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

Usual preparation

- Glass ampoule containing 50 mg of tenecteplase, in powder form with a pre-filled syringe containing 10 ml of sterile water.

Common interactions

- None.

More Information

Version history

11 September 2023 1.0.4

Update: Indication wording has been changes. Cautions and controindications have been updated.